

m -reflexive qh dgas

Abstract

We study exceptional standard objects in the m -extended module category whose left-dual objects have cohomology in degrees $0, \dots, m-1$. The definition specialises to classical quasi-heredity for $m=1$. For every ordering of the vertices of a finite acyclic graded quiver, we construct suitable standard dg modules and prove fullness in both $\text{per } A$ and $\mathcal{D}_{\text{fd}}(A)$. The construction allows a non-zero differential; its cohomological bounds are controlled by the degrees of paths.

1 Definition and classical specialisation

Fix an algebraically closed field $\mathbb{k}$ and an integer $m \geq 1$. All dg modules are right modules, and all complexes are cochain complexes: $d_X : X^q \rightarrow X^{q+1}$ and $H^q(X[s]) = H^{q+s}(X)$. Let A be a proper connective dg algebra satisfying

$$\dim_{\mathbb{k}} H^*(A) < \infty, \quad H^q(A) = 0 \quad (q \notin [1-m, 0]). \quad (1.1)$$

Assume that $H^0(A)$ is basic and that closed degree-zero orthogonal idempotents $e_1, \dots, e_n$ sum to 1, with classes forming a complete set of primitive idempotents of $H^0(A)$. Set

$$\begin{aligned} P_v &= e_v A, & \mathcal{D}_{\text{fd}}(A) &= \{X \in \mathcal{D}(A) : \dim_{\mathbb{k}} H^*(X) < \infty\}, \\ \text{per } A &= \text{thick}_{\mathcal{D}(A)}(A), & \mathcal{H}_A^m &= \mathcal{D}_{\text{fd}}(A)^{[1-m, 0]}. \end{aligned} \quad (1.2)$$

For a class $\mathcal{S} \subseteq \mathcal{D}_{\text{fd}}(A)$, $\text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\mathcal{S})$ denotes closure under shifts, cones, and direct summands. Properness gives $\text{per } A \subseteq \mathcal{D}_{\text{fd}}(A)$.

The category $\mathcal{H}_A^m$ is extriangulated, with

$$\mathbb{E}_{\mathcal{H}_A^m}(X, Y) = \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(X, Y[1]), \quad \mathcal{P}(\mathcal{H}_A^m) = \text{add } A \quad (X, Y \in \mathcal{H}_A^m) \quad (1.3)$$

[MP25, Corollary 2.3]. A conflation $K \rightarrow P \rightarrow X \dashrightarrow$ in $\mathcal{H}_A^m$ is induced by a triangle $K \rightarrow P \rightarrow X \rightarrow K[1]$ in $\mathcal{D}_{\text{fd}}(A)$ with $K, P, X \in \mathcal{H}_A^m$. We write $\text{Filt}_{\mathcal{H}_A^m}(\mathcal{S})$ for the finite extension closure of $\mathcal{S}$ in $\mathcal{H}_A^m$, including zero and finite direct sums.

An object $E \in \mathcal{D}_{\text{fd}}(A)$ is *exceptional* if

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(E, E[t]) \cong \begin{cases} \mathbb{k}, & t = 0, \\ 0, & t \neq 0 \end{cases} \quad (t \in \mathbb{Z}). \quad (1.4)$$

An ordered tuple $(E_1, \dots, E_n)$ of exceptional objects is an *exceptional sequence* if

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(E_i, E_j[t]) = 0 \quad (1 \leq j < i \leq n, t \in \mathbb{Z}). \quad (1.5)$$

The indices i, j in (1.5) denote positions in the sequence. For exceptional objects $\Delta_1, \dots, \Delta_n$, a permutation $\sigma \in S_n$ lists the vertex labels in the standard sequence

$$\Delta^\sigma = (\Delta_{\sigma(1)}, \dots, \Delta_{\sigma(n)}). \quad (1.6)$$

Thus $\sigma^{-1}(v)$ is the position of Δ_v in Δ^σ . Define the total order $\leq_\sigma$ on the vertex labels $\{1, \dots, n\}$ by

$$u \leq_\sigma v \iff \sigma^{-1}(u) \leq \sigma^{-1}(v). \quad (1.7)$$

We write $u <_\sigma v$ if $u \leq_\sigma v$ and $u \neq v$. The least and greatest weights are $\sigma(1)$ and $\sigma(n)$, respectively. Thus smaller weights occur earlier in (1.6). By (1.5), the sequence Δ^σ is exceptional precisely when

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_w, \Delta_v[t]) = 0 \quad (v <_\sigma w, t \in \mathbb{Z}). \quad (1.8)$$

The *left-dual sequence* of Δ^σ is an exceptional sequence $(\nabla_{\sigma(n)}, \dots, \nabla_{\sigma(1)})$, listed in decreasing order for $\leq_\sigma$, whose terms satisfy

$$\nabla_v \in \text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_1, \dots, \Delta_n) \quad (1 \leq v \leq n) \quad (1.9)$$

and whose pairing with the standard objects is

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_u, \nabla_v[t]) \cong \begin{cases} \mathbb{k}, & u = v, t = 0, \\ 0, & \text{otherwise} \end{cases} \quad (1 \leq u, v \leq n, t \in \mathbb{Z}). \quad (1.10)$$

We use the convention of [BB26, §2.3]. The objects ∇_v , when they exist, are determined up to isomorphism in $\text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_1, \dots, \Delta_n)$ by (1.10).

Definition 1.1 (*m-reflexive qh dga*). An algebra A satisfying (1.1) is an *m-reflexive qh dga with respect to σ* if there exist objects $\Delta_v \in \mathcal{H}_A^m$ such that:

(R1) Δ^σ satisfies (1.4) and (1.8);

(R2) for each vertex v , there is a conflation

$$K_v \longrightarrow P_v \xrightarrow{p_v} \Delta_v \dashrightarrow \quad \text{in } \mathcal{H}_A^m; \quad (1.11)$$

(R3) the left-dual sequence of Δ^σ exists, with

$$\nabla_v \in \mathcal{D}_{\text{fd}}(A)^{[0, m-1]} \quad (1 \leq v \leq n); \quad (1.12)$$

(R4) Δ^σ is full:

$$\text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_1, \dots, \Delta_n) = \mathcal{D}_{\text{fd}}(A). \quad (1.13)$$

In Definition 1.1, Δ^σ is the chosen exceptional sequence; condition R3 concerns the left-dual objects ∇_v specified by (1.10). Equation (1.10) implies $H^0(\Delta_v) \neq 0$: otherwise $\Delta_v \in \mathcal{D}_{\text{fd}}(A)^{\leq -1}$ and $\nabla_v \in \mathcal{D}_{\text{fd}}(A)^{\geq 0}$ would give $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_v, \nabla_v) = 0$. The epimorphism $H^0(p_v) : H^0(P_v) \rightarrow H^0(\Delta_v)$ is therefore a projective cover. In particular, the top of $H^0(\Delta_v)$ is the simple $H^0(A)$ -module at vertex v .

Proposition 1.2 (Classical specialisation). *Suppose $H^q(A) = 0$ for $q \neq 0$, and put $B = H^0(A)$. The following conditions are equivalent:*

- (i) A is a 1-reflexive qh dga with respect to σ ;
- (ii) B is quasi-hereditary with respect to σ .

Proof. The quasi-isomorphism $A \simeq B$ identifies $\mathcal{H}_A^1$ with $\text{mod } B$.

(ii) $\Rightarrow$ (i). The classical standard modules have projective-cover presentations $P_v \twoheadrightarrow \Delta_v$ whose kernels are filtered by Δ_w with $v <_\sigma w$ [DR92]. The classical standard and costandard sequences are full exceptional sequences satisfying (1.10) [BB26, Lemma 2.2]. The classical projective-cover presentations and standard and costandard sequences satisfy conditions R1–R4.

(i) $\Rightarrow$ (ii). Conditions **R1**, **R3**, and **R4** give full exceptional sequences $(\Delta_{\sigma(1)}, \dots, \Delta_{\sigma(n)})$ and $(\nabla_{\sigma(n)}, \dots, \nabla_{\sigma(1)})$ in $\mathcal{D}^b(\text{mod } B)$, with all terms in $\text{mod } B$. By [BB26, Theorem 3.8], the sequences Δ^σ and $(\nabla_{\sigma(n)}, \dots, \nabla_{\sigma(1)})$ determine a highest weight structure on $\text{mod } B$. The projective covers $H^0(p_v)$ identify the highest weight structure's standard-object labels with the vertices ordered by σ . $\square$

Example 1.3 (The order for two classical standards). Take $A = B = \mathbb{k}(1 \xrightarrow{a} 2)$ concentrated in degree zero, with $e_1 a = a = a e_2$, $m = 1$, and $\sigma = \text{id}$. Thus $1 <_\sigma 2$. Let S_v be the simple top of $P_v = e_v B$, for $v = 1, 2$. Then

$$\Delta_1 = S_1, \quad \Delta_2 = S_2 = P_2, \quad \nabla_1 = S_1, \quad \nabla_2 = P_1. \quad (1.14)$$

The projective presentation

$$0 \longrightarrow S_2 \longrightarrow P_1 \longrightarrow S_1 \longrightarrow 0 \quad (1.15)$$

gives

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(B)}(S_2, S_1[t]) = 0 \quad (t \in \mathbb{Z}), \quad \text{Hom}_{\mathcal{D}_{\text{fd}}(B)}(S_1, S_2[1]) \cong \mathbb{k}. \quad (1.16)$$

Consequently $(\Delta_1, \Delta_2) = (S_1, S_2)$ is full exceptional, whereas (Δ_2, Δ_1) is not exceptional. The left-dual sequence of (Δ_1, Δ_2) specified by (1.10) is $(\nabla_2, \nabla_1) = (P_1, S_1)$.

2 Graded path dg algebras

Let Q be a finite acyclic quiver with vertex set $Q_0 = \{1, \dots, n\}$, where $n \geq 1$. Give each arrow an integer degree at most 0. For a path p , let $s(p)$, $t(p)$, and $|p|$ denote its source, target, and degree. Trivial paths have degree 0. We concatenate paths from left to right, so $e_u a = a = a e_v$ for an arrow $a : u \rightarrow v$. Let

$$A = (\mathbb{k}Q, d_A), \quad d_A : A^q \longrightarrow A^{q+1}, \quad d_A^2 = 0, \quad (2.1)$$

where d_A is a graded derivation. Since $A^1 = 0$, all e_v are closed. Define

$$M = 1 - \min\{|p| : p \text{ is a path in } Q\}. \quad (2.2)$$

Thus $A^q = 0$ for $q \notin [1 - M, 0]$.

For complexes of $\mathbb{k}$ -vector spaces, write $\mathbb{D}V = \text{Hom}_{\mathbb{k}}^\bullet(V, \mathbb{k})$, with

$$(\mathbb{D}V)^q = \text{Hom}_{\mathbb{k}}(V^{-q}, \mathbb{k}), \quad d_{\mathbb{D}V}(\varphi) = (-1)^{q+1} \varphi d_V \quad (\varphi \in (\mathbb{D}V)^q).$$

For a left dg A -module V , the right action on $\mathbb{D}V$ is $(\varphi a)(x) = \varphi(ax)$. For right dg modules X, Y , let $\mathcal{H}om_A(X, Y)$ be the cochain complex of homogeneous A -linear maps, with

$$d(f) = d_Y f - (-1)^q f d_X \quad (f \in \mathcal{H}om_A(X, Y)^q).$$

We write $\mathbf{R}\mathcal{H}om_A(X, Y)$ for the corresponding derived Hom complex.

Theorem 2.1 (Standards for every ordering). *Let A be the path dga (2.1), and fix $\sigma \in S_n$. For each vertex v , set*

$$\begin{aligned} e_{>v}^\sigma &= \sum_{v <_\sigma w} e_w, & B_v &= A/Ae_{>v}^\sigma A, & K_v &= e_v A e_{>v}^\sigma A, \\ \Delta_v &= e_v B_v = P_v/K_v, & C_v &= B_v e_v, & \nabla_v &= \mathbb{D}C_v. \end{aligned} \quad (2.3)$$

Then A is an M -reflexive qh dga with respect to σ , with standard objects Δ_v , left-dual objects ∇_v , and conflations induced by

$$0 \longrightarrow K_v \longrightarrow P_v \xrightarrow{p_v} \Delta_v \longrightarrow 0. \quad (2.4)$$

Moreover,

$$\text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_1, \dots, \Delta_n) = \text{per } A = \mathcal{D}_{\text{fd}}(A). \quad (2.5)$$

The objects in (2.3) also satisfy Definition 1.1 for every $m \geq M$.

Proof. The ideals $Ae_{>v}^\sigma A$ are dg ideals, because the idempotents e_w are closed. Thus every object in (2.3) is a dg module, and (2.4) is a short exact sequence of dg modules.

1. Finite projective filtrations of the kernels. Let $\mathcal{F}_v$ be the set of paths $p = (v = v_0 \rightarrow v_1 \rightarrow \cdots \rightarrow v_\ell)$ such that

$$v <_\sigma v_\ell, \quad v_j \leq_\sigma v \quad (0 \leq j < \ell).$$

Every basis path of K_v has a unique prefix in $\mathcal{F}_v$. As graded right modules,

$$K_v = \bigoplus_{p \in \mathcal{F}_v} pA \cong \bigoplus_{p \in \mathcal{F}_v} P_{t(p)}[-|p|]. \quad (2.6)$$

The summands pA need not be dg submodules. Write the unique expansion

$$\begin{aligned} d_A(p) &= \sum_{p' \in \mathcal{F}_v} p' a_{p'p}, \\ a_{p'p} &\in e_{t(p')} A^{|p|+1-|p'|} e_{t(p)}. \end{aligned} \quad (2.7)$$

Every non-zero homogeneous coefficient $a_{p'p}$ satisfies $|p'| = |p| + 1 - |a_{p'p}| \geq |p| + 1$. Consequently,

$$F^h K_v = \bigoplus_{\substack{p \in \mathcal{F}_v \\ |p| \geq h}} pA, \quad F^h K_v / F^{h+1} K_v \cong \bigoplus_{\substack{p \in \mathcal{F}_v \\ |p| = h}} P_{t(p)}[-h] \quad (2.8)$$

is a finite filtration by dg submodules, with $h \in \mathbb{Z}$. Indeed, modulo $F^{h+1} K_v$, $d_A(px) = (-1)^h p d_A(x)$ for $|p| = h$, which is the differential on $P_{t(p)}[-h]$. Each vertex $t(p)$ satisfies $v <_\sigma t(p)$.

Filtration (2.8) makes K_v perfect and homotopically projective: $\mathcal{H}om_A(K_v, -)$ preserves quasi-isomorphisms. Sequence (2.4) then makes Δ_v perfect. Descending induction in $<_\sigma$, using (2.8) and (2.4), gives

$$P_v \in \text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_v, \Delta_w : v <_\sigma w), \quad \text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_1, \dots, \Delta_n) = \text{per } A. \quad (2.9)$$

2. Exceptionality of the standard sequence. Suppose $v \leq_\sigma u$. Every projective factor of K_u in (2.8) is a shift of P_w with $u <_\sigma w$. Since $\Delta_v e_w = 0$,

$$\mathbf{R}\mathcal{H}om_A(K_u, \Delta_v) = 0, \quad \mathbf{R}\mathcal{H}om_A(P_u, \Delta_v) \simeq \Delta_v e_u = \begin{cases} \mathbb{k}, & u = v, \\ 0, & v <_\sigma u. \end{cases}$$

The triangle induced by (2.4) yields (1.4) for Δ_v and (1.8) for Δ^σ .

3. Fullness in $\mathcal{D}_{\text{fd}}(A)$. Let $J \subseteq A$ be the ideal generated by all arrows. Acyclicity implies $d_A(J) \subseteq J$. For each vertex v , define the simple dg module $S_v = P_v / e_v J$, concentrated in degree 0. The outgoing arrows form a graded free basis of $e_v J$ over the corresponding projectives. The generator-degree argument of (2.7)–(2.8) therefore gives a finite projective filtration of $e_v J$. The exact sequence $0 \rightarrow e_v J \rightarrow P_v \rightarrow S_v \rightarrow 0$ makes every S_v perfect. The modules $S_1, \dots, S_n$ are the simple $H^0(A)$ -modules. Finite-length filtrations and cohomological truncations give

$$\mathcal{D}_{\text{fd}}(A) = \text{thick}_{\mathcal{D}_{\text{fd}}(A)}(S_1, \dots, S_n) \subseteq \text{per } A.$$

Properness and (2.9) prove (2.5).

4. The specified left-dual objects. For vertices u, v , restriction along $K_u \hookrightarrow P_u$ defines a cochain map

$$\rho_{uv} : \mathcal{H}om_A(P_u, \nabla_v) \longrightarrow \mathcal{H}om_A(K_u, \nabla_v). \quad (2.10)$$

If $u <_\sigma v$, a basis of $e_u C_v$ consists of paths $q : u \rightsquigarrow v$ all of whose vertices are $\leq_\sigma v$. Each such q has a unique factorisation $q = ps$ with $p \in \mathcal{F}_u$. Write $q^\vee$ and $s^\vee$ for the corresponding basis functionals. The homogeneous A -linear map $f_q \in \mathcal{H}om_A(P_u, \nabla_v)^{-|q|}$ determined by $f_q(e_u) = q^\vee$ satisfies

$$\rho_{uv}(f_q)(p') = \begin{cases} s^\vee, & p' = p, \\ 0, & p' \neq p \end{cases} \quad (p' \in \mathcal{F}_u). \quad (2.11)$$

Conversely, every generator-assignment basis vector of $\mathcal{H}om_A(K_u, \nabla_v)$ equals $\rho_{uv}(f_q)$ for a unique path q belonging to the stated basis of $e_u C_v$. Hence ρ_{uv} is an isomorphism of cochain complexes.

If $v <_\sigma u$, both complexes in (2.10) vanish. If $u = v$, the source of (2.10) is $\mathbb{k}$ in degree 0 and its target is zero. Applying $\mathbf{R}\mathcal{H}om_A(-, \nabla_v)$ to (2.4), using the homotopical projectivity of P_u and K_u , proves (1.10).

To verify exceptionality of $(\nabla_{\sigma(n)}, \dots, \nabla_{\sigma(1)})$, apply Steps 1–3 to the graded opposite algebra A^{op} . The left dg modules $C_v = B_v e_v$ are the resulting standard objects. Thus $(C_{\sigma(1)}, \dots, C_{\sigma(n)})$ is exceptional in $\mathcal{D}_{\text{fd}}(A^{\text{op}})$. Finite-dimensional vector-space duality gives

$$\mathcal{H}om_{\mathcal{D}_{\text{fd}}(A)}(\nabla_u, \nabla_v[t]) \cong \mathcal{H}om_{\mathcal{D}_{\text{fd}}(A^{\text{op}})}(C_v, C_u[t]).$$

The sequence $(\nabla_{\sigma(n)}, \dots, \nabla_{\sigma(1)})$ is therefore exceptional. Equation (2.5) places every ∇_v in $\text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_1, \dots, \Delta_n)$.

5. Cohomological bounds and deflations. The modules P_v, K_v, Δ_v , and C_v have graded components only in $[1 - M, 0]$, by (2.2) and (2.3). Therefore

$$P_v, K_v, \Delta_v \in \mathcal{H}_A^M, \quad \nabla_v = \mathbb{D}C_v \in \mathcal{D}_{\text{fd}}(A)^{[0, M-1]}.$$

Sequence (2.4) induces a conflation in $\mathcal{H}_A^M$, so p_v is a deflation. Conditions R1–R4 follow from Steps 2–5. For $m \geq M$, both cohomological windows only enlarge. □

Corollary 2.2 (Zero differential). *If $d_A = 0$, then M in (2.2) is the smallest integer for which A satisfies (1.1). Every ordering σ therefore gives an M -reflexive qh dga.*

Proof. For $d_A = 0$, a path of minimal degree represents a non-zero class in $H^{1-M}(A)$. Apply Theorem 2.1. □

Proposition 2.3 (Differentials on tree quivers). *If the underlying unoriented graph of Q is a tree, then the differential d_A in (2.1) is zero.*

Proof. For an arrow $a : u \rightarrow v$, closure of e_u, e_v gives $d_A(a) \in e_u A^{|a|+1} e_v$. The only path from u to v is a , so $e_u A e_v = \mathbb{k}a$ is concentrated in degree $|a|$. Thus $d_A(a) = 0$ for every arrow a ; the derivation d_A vanishes. □

3 Examples

In Examples 3.1–3.3, the objects Δ_v, K_v, C_v , and ∇_v are defined by (2.3). The symbol S_v denotes the simple module at vertex v , as in Step 3 of the proof of Theorem 2.1. All quotient modules inherit the indicated path actions.

Example 3.1 (A graded quiver of type A_4). Let

$$Q : 1 \xrightarrow{a} 2 \xrightarrow{b} 3 \xrightarrow{c} 4, \quad |a| = |b| = |c| = -1, \quad d_A = 0.$$

The path abc has degree -3 , so $M = 4$. Choose $2 <_\sigma 4 <_\sigma 1 <_\sigma 3$, and put $U = P_1/(abA)$, with basis e_1, a . Formula (2.3) gives

v	Δ_v	K_v	∇_v
1	U	$abA \cong P_3[2]$	S_1
2	S_2	$bA \cong P_3[1]$	S_2
3	P_3	0	$\mathbb{D}(Ae_3)$
4	S_4	0	S_4

(3.1)

The module Ae_3 has basis e_3, b, ab in degrees $0, -1, -2$. Hence

$$\begin{aligned} H^0(U) = S_1, \quad H^{-1}(U) = S_2, \quad H^0(P_3) = S_3, \quad H^{-1}(P_3) = S_4, \\ H^0(\nabla_3) = S_3, \quad H^1(\nabla_3) = S_2, \quad H^2(\nabla_3) = S_1. \end{aligned} \quad (3.2)$$

Theorem 2.1 gives the full exceptional sequence (S_2, S_4, U, P_3) with left-dual sequence $(\nabla_3, S_1, S_4, S_2)$. The presentations of U and S_2 are the conflations

$$P_3[2] \xrightarrow{x \mapsto abx} P_1 \longrightarrow U \dashrightarrow, \quad P_3[1] \xrightarrow{x \mapsto bx} P_2 \longrightarrow S_2 \dashrightarrow \quad \text{in } \mathcal{H}_A^4.$$

Example 3.2 (A graded quiver of type D_4). Let

$$Q : 1 \xrightarrow{a} 2 \xrightarrow{b} 3, \quad 2 \xrightarrow{c} 4, \quad |a| = |c| = -1, \quad |b| = 0, \quad d_A = 0.$$

The path ac has degree -2 , so $M = 3$. Choose $1 <_\sigma 3 <_\sigma 2 <_\sigma 4$, and set $V = P_2/(cA)$, with basis e_2, b in degree 0. Then

v	Δ_v	K_v	∇_v
1	S_1	$aA \cong P_2[1]$	S_1
2	V	$cA \cong P_4[1]$	$\mathbb{D}(Ae_2)$
3	S_3	0	S_3
4	S_4	0	$\mathbb{D}(Ae_4)$

(3.3)

The left modules Ae_2 and Ae_4 have respective bases e_2, a and e_4, c, ac . Consequently,

$$\begin{aligned} H^0(\nabla_2) = S_2, \quad H^1(\nabla_2) = S_1, \\ H^0(\nabla_4) = S_4, \quad H^1(\nabla_4) = S_2, \quad H^2(\nabla_4) = S_1. \end{aligned} \quad (3.4)$$

The standard sequence (S_1, S_3, V, S_4) and its left-dual sequence $(\nabla_4, \nabla_2, S_3, S_1)$ satisfy Definition 1.1 with $m = 3$. For example, $H^{-1}(K_1) = V$ and $H^{-2}(K_1) = S_4$, so the conflation $K_1 \rightarrow P_1 \rightarrow S_1 \dashrightarrow$ belongs to $\mathcal{H}_A^3$.

Example 3.3 (A non-zero differential and the choice of width). Take

$$\begin{aligned} Q : 1 \xrightarrow{a} 2 \xrightarrow{b} 3 \xrightarrow{h} 4, \quad c : 1 \rightarrow 3, \\ |a| = |b| = 0, \quad |c| = |h| = -1, \quad d_A(c) = ab, \quad d_A(a) = d_A(b) = d_A(h) = 0. \end{aligned} \quad (3.5)$$

The only non-zero path differentials are $d_A(c) = ab$ and $d_A(ch) = abh$. Thus

$$H^0(A) = \mathbb{k}\{e_1, e_2, e_3, e_4, a, b\}, \quad H^{-1}(A) = \mathbb{k}\{h, bh\}, \quad H^q(A) = 0 \quad (q \notin [-1, 0]). \quad (3.6)$$

The smallest parameter allowed by (1.1) is $m = 2$, whereas the path ch gives $M = 3$ in (2.2).

Choose $3 <_\sigma 4 <_\sigma 1 <_\sigma 2$, and put $X = P_1/(aA)$. The quotient X has basis e_1, c, ch and zero differential. Formula (2.3) gives

$$\begin{aligned} (\Delta_{\sigma(1)}, \Delta_{\sigma(2)}, \Delta_{\sigma(3)}, \Delta_{\sigma(4)}) &= (S_3, S_4, X, P_2), \\ (K_1, K_2, K_3, K_4) &= (aA, 0, hA, 0), \\ H^0(X) &= S_1, \quad H^{-1}(X) = S_3, \quad H^{-2}(X) = S_4. \end{aligned} \quad (3.7)$$

The objects in (3.7), with the objects ∇_v of (2.3), define a 3-reflexive qh dga by Theorem 2.1. The same standard X does not belong to $\mathcal{H}_A^2$, because the class of ch generates $H^{-2}(X)$.

A different order isolates the condition on the left-dual objects. Choose $1 <_\tau 3 <_\tau 4 <_\tau 2$, and denote the objects of (2.3) for τ by $\Delta_v^\tau, K_v^\tau, C_v^\tau$, and ∇_v^τ . Then

$$\begin{aligned} (\Delta_1^\tau, \Delta_2^\tau, \Delta_3^\tau, \Delta_4^\tau) &= (S_1, P_2, S_3, S_4), \\ C_4^\tau &= (A/Ae_2A)e_4 = \mathbb{k}\{e_4, h, ch\}, \quad d_{C_4^\tau} = 0, \\ H^0(\nabla_4^\tau) &= S_4, \quad H^1(\nabla_4^\tau) = S_3, \quad H^2(\nabla_4^\tau) = S_1. \end{aligned} \tag{3.8}$$

Every Δ_v^τ and K_v^τ belongs to $\mathcal{H}_A^2$. Indeed, $K_1^\tau = \mathbb{k}\{a, ab, abh, c, ch\}$ has cohomology S_2 in degree 0, $K_3^\tau = hA \cong P_4[1]$, and $K_2^\tau = K_4^\tau = 0$. However, ∇_4^τ in (3.8) violates (1.12) for $m = 2$. Theorem 2.1 applies to τ with $m = 3$.

Equations (3.7) and (3.8) concern the specific construction (2.3); they do not exclude other standard families for σ or τ at width 2.

4 Questions

Theorem 2.1 settles existence for graded path dg algebras at every width $m \geq M$. For a general algebra satisfying (1.1), several structural questions remain.

Foundational questions

- F1. Generation and perfection.** If conditions R1–R3 hold, when does (1.13) follow? For an m -reflexive qh dga, when do the standard objects Δ_v belong to $\text{per } A$?
- F2. A prescribed cohomological width.** Characterise the orders admitting Definition 1.1 for a fixed m satisfying (1.1). In Example 3.3, determine whether standard families other than (2.3) exist at width 2 for the orders σ and τ .
- F3. Filtrations by shifted standards.** Determine hypotheses under which the presentations (1.11) can be chosen with

$$K_v \in \text{Filt}_{\mathcal{H}_A^m} \{ \Delta_w[s] : v <_\sigma w, 0 \leq s < m, \Delta_w[s] \in \mathcal{H}_A^m \}. \tag{4.1}$$

Establish the relation between (4.1) and the cohomological requirement (1.12).

Further directions

- P1. Heredity and recollement.** Relate Definition 1.1 to filtrations by dg ideals and recollements whose successive factors are equivalent to $\mathcal{D}^b(\text{mod } \mathbb{k})$.
- P2. Standardisation and Ringel duality.** Study the extension-closed subcategories $\text{Filt}_{\mathcal{H}_A^m} \{ \Delta_v \}$ and $\text{Filt}_{\mathcal{H}_A^m} \{ \nabla_v[m-1] \}$. Determine conditions producing a tilting or silting object T and an m' -reflexive qh structure on $\mathbf{R}\text{End}_A(T)$; determine the resulting integer $m' \geq 1$.
- P3. Reciprocity and dimension bounds.** Seek a graded form of BGG reciprocity and homological bounds in terms of m and n . In the Grothendieck group $K_0(\mathcal{D}_{\text{fd}}(A))$, write $[Y]$ for the class of $Y \in \mathcal{D}_{\text{fd}}(A)$. Filtrations in (4.1) must account for $[Y[s]] = (-1)^s[Y]$ for $s \in \mathbb{Z}$.

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